

IS 10262-Based Comparative Mix Design and Sustainability Assessment of M25 OPC and LC3 Concrete

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Abstract

This desk study proportions an M25 concrete per IS 10262:2019 using ordinary Portland cement (OPC) and designs a limestone calcined clay cement (LC3-50: 50% clinker, 30% calcined clay, 15% limestone, 5% gypsum) counterpart with the same binder content (413 kg/m³) and water content (186 kg/m³). The LC3 mix is volumetrically corrected for the lower specific gravity of the blended binder. Embodied CO₂ is estimated on a ground-to-gate basis using Indian emission factors published by IIT Madras, and binder raw-material cost is estimated using Indian input prices from an IIT Delhi economic analysis of LC3. The LC3 mix is estimated to embody **236 kg CO₂-eq/m³** against **346** for OPC (**32% lower**), with a 31–41% lower binder raw-material cost. Strength is not measured here; published data are reviewed instead, and they indicate that equal-binder LC3 mixes can show lower 28-day strength than OPC while the gap narrows with age. Trial batches are therefore required before the LC3 proportions can be adopted for M25.

1. Introduction

Cement production accounts for roughly 7–8% of global anthropogenic CO₂, and most of it comes from producing clinker, both from fuel combustion and from decomposing limestone. Lowering the clinker content of cement is therefore the most direct lever for decarbonising concrete. India's usual clinker substitutes, fly ash and slag, are limited in quality and local availability. Limestone calcined clay cement (LC3) replaces about half of the clinker with calcined kaolinitic clay and crushed limestone, which are abundant; the metakaolin in the clay reacts pozzolanically, and the limestone reacts with the aluminates to form carboaluminates, which is why strength can be retained at 50% clinker [4, 5]. LC3 has been produced in pilot trials in India [10, 11] and is now standardised as IS 18189:2023, Portland Calcined Clay Limestone Cement [3].

Objective. To quantify, for a common structural grade (M25), how substituting OPC with LC3-50 changes the mix proportions, embodied CO₂ and binder cost, and to identify from published data what that substitution implies for strength. **Scope.** This is a calculation and literature study. No specimens were cast or tested. Every value taken from published work is identified as secondary data.

2. Methodology

2.1 Design basis

Parameter	Value	Basis
Grade / characteristic strength	M25 / 25 MPa	Design choice
Exposure	Moderate	IS 456:2000 Table 5: max w/c 0.50, min. cement 300 kg/m ³
Max. nominal aggregate size	20 mm	Design choice
Workability (slump)	50 mm	Design choice
Fine aggregate	Zone II	Assumed
Specific gravities (OPC / FA / CA)	3.15 / 2.60 / 2.70	Assumed typical values; to be replaced with test data
Entrapped air	2% (0.02 m ³)	IS 10262:2019, 20 mm aggregate
LC3 composition	50 : 30 : 15 : 5	Clinker : calcined clay : limestone : gypsum [4, 8]

2.2 OPC mix proportioning (IS 10262:2019)

Step 1 — Target mean strength. With $\sigma = 4.0$ MPa for M20–M25 (IS 10262 Table 2):

$$f_{ck} = f_{ck} + 1.65\sigma = 25 + 1.65 \times 4 = 31.6 \text{ MPa}$$

Step 2 — Water-cement ratio. 0.45 was adopted. It is below the IS 456 durability limit of 0.50 for moderate exposure and is a conservative choice for a 31.6 MPa target; it should be confirmed against the strength-w/c relationship in IS 10262 (Fig. 1) for the cement grade actually used.

Step 3 — Water content. 186 kg/m³ for 20 mm aggregate at 50 mm slump (IS 10262 Table 4).

Step 4 — Cement content. $C = 186 / 0.45 = 413.3$, taken as 413 kg/m³. This is above the 300 kg/m³ minimum and below the 450 kg/m³ maximum of IS 456.

Step 5 — Aggregate volume. $V_{agg} = 1 - (V_{cement} + V_{water} + V_{air}) = 1 - (0.1311 + 0.1860 + 0.0200) = 0.6629 \text{ m}^3$.

Step 6 — Coarse/fine split. IS 10262 Table 5 gives a coarse-aggregate volume fraction of 0.62 for 20 mm aggregate with Zone II sand at w/c 0.50; it is increased by 0.01 for each 0.05 decrease in w/c, giving 0.63. Hence CA = $0.6629 \times 0.63 \times 2.70 \times 1000 = 1128 \text{ kg/m}^3$ and FA = $0.2453 \times 2.60 \times 1000 = 638 \text{ kg/m}^3$. The resulting mix is **1 : 1.54 : 2.73** with w/c 0.45.

2.3 LC3-50 mix

The binder mass (413 kg/m³), free water and coarse-aggregate fraction were held constant, so that binder chemistry is the main variable. The blended binder's specific gravity was computed as a mass-weighted harmonic mean of its constituents (clinker 3.15, calcined clay 2.67 and limestone 2.63 [6], gypsum 2.32), giving 2.86. Because the LC3 binder occupies more volume (0.1444 m³ vs 0.1311 m³), the aggregate masses were reduced slightly so that the mix still yields exactly 1 m³. This is a *theoretical equal-water comparison*: calcined clay has a high specific surface and usually raises water demand, so a real LC3 mix would need a superplasticiser dose adjusted to reach the same slump [6, 7].

2.4 Embodied CO₂

The system boundary is ground-to-gate (raw-material extraction to material at the gate). Transport, superplasticiser, batching and curing are excluded. Emission factors were taken from the life-cycle inventory of Pillai et al. [6], which uses Indian cement-plant data (Table 1). **Scenario A** (main result) applies the published cement-level factors. **Scenario B** (cross-check) rebuilds the binder factor from its constituents; because it omits grinding electricity, it gives an upper bound on the saving.

Material	kg CO ₂ -eq / kg	Use
OPC (cement level)	0.82	Scenario A
LC3 (cement level)	0.554	Scenario A
Clinker	0.85	Scenario B
Calcined clay (calcination)	0.127	Scenario B
Limestone	0.00219	Scenario B
Gypsum	0	Scenario B
Fine and coarse aggregates	0.00419	Both
Water	0.000658	Both

Table 1. Ground-to-gate emission factors from Pillai et al. (2019) [6], Table 3.

2.5 Binder cost

Market prices for LC3 are not yet established, so the cost of binder raw materials was estimated from the Indian input prices used by Joseph and Bishnoi [8] for a grinding-unit scenario: clinker ₹6,000/t, gypsum ₹1,800/t, limestone ₹200/t and calcined clay ₹1,000–3,000/t. OPC was taken as 95% clinker and 5% gypsum. These are 2016 prices and exclude grinding energy, capital cost, taxes and margins, so the results indicate relative cost, not selling price.

2.6 Literature data

Strength data were taken from two sources that report OPC and LC3 side by side: Indian concrete data from IIT Madras [6] and a mortar study that varies the replacement level [9]. They are presented separately from this study's calculations.

3. Results

3.1 Mix proportions

Constituent (kg/m ³)	OPC	LC3-50
Clinker	392.3	206.5
Calcined clay	—	123.9
Limestone	—	61.9
Gypsum	20.7	20.7
Total binder	413	413
Water	186	186
Fine aggregate	638	625
Coarse aggregate	1128	1105
Water-binder ratio	0.45	0.45
Mix ratio (binder : FA : CA)	1 : 1.54 : 2.73	1 : 1.51 : 2.68

Table 2. Calculated mix proportions per m³ (this study).

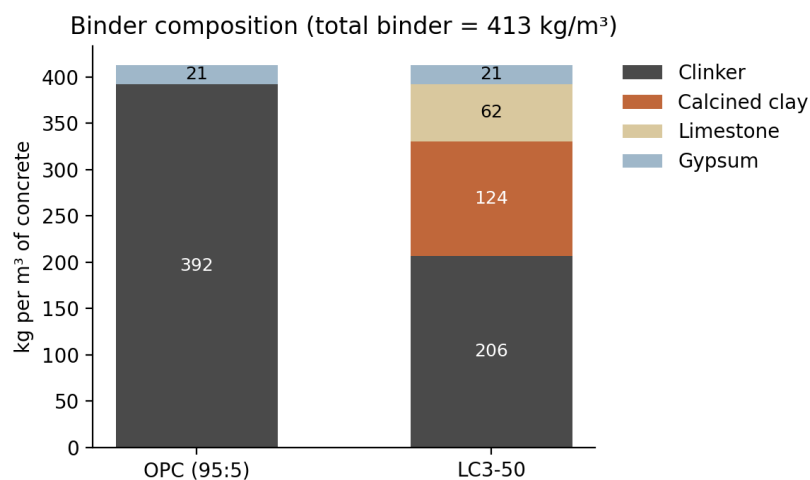


Figure 1. Binder composition. LC3-50 contains 47% less clinker than OPC at the same binder mass.

3.2 Embodied CO₂

kg CO ₂ -eq / m ³	OPC	LC3-50
Binder	338.7	228.8
Aggregates	7.4	7.2
Water	0.12	0.12
Total (Scenario A)	346	236

Table 3. Embodied CO₂ per m³. Reduction with LC3: 31.8% (Scenario A).

The binder contributes about 97–98% of the total in both mixes, so the result depends almost entirely on the binder factor. Scenario B gives binder factors of 0.807 (OPC) and 0.463 (LC3) kg CO₂-eq/kg, a 43% reduction at binder level. The Scenario A result (32%) sits within the 30–40% range commonly reported for LC3 cement, and the difference from Scenario B reflects the processing energy that Scenario B omits.

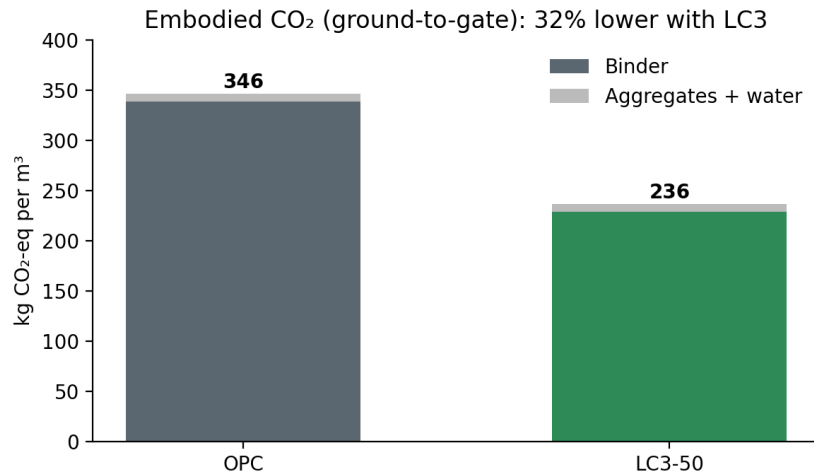


Figure 2. Embodied CO₂ per m³ of concrete (ground-to-gate, Scenario A).

3.3 Binder cost

Binder	₹ / tonne binder	₹ / m ³ concrete	Saving vs OPC
OPC	5,790	2,391	—
LC3-50, clay ₹1,000/t	3,420	1,412	41%
LC3-50, clay ₹2,000/t	3,720	1,536	36%
LC3-50, clay ₹3,000/t	4,020	1,660	31%

Table 4. Binder raw-material cost (2016 input prices from [8]; excludes grinding, capex, taxes).

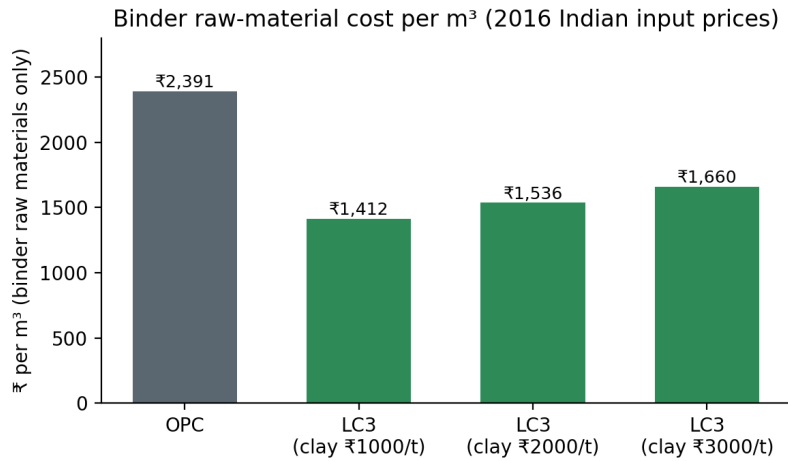


Figure 3. Binder raw-material cost per m³ for three calcined-clay prices.

4. Strength: review of published data

Table 5 reproduces 28-day cube strengths and embodied CO₂ from the IIT Madras study [6], which designed OPC, fly-ash (PFA) and LC3 concretes in two strength classes. The CO₂-per-MPa column was calculated for this report from the published values.

Mix	Binder	Binder (kg/m ³)	w/b	f ₂₈ (MPa)	CO ₂ (kg/m ³)	CO ₂ per MPa
OPC-1	OPC	310	0.50	45.4	335	7.38
LC3-1S	LC3	310	0.50	38.8	255	6.57
LC3-1P	LC3	360	0.45	40.7	280	6.88
PFA-1	OPC+30% fly ash	360	0.45	37.6	290	7.71
OPC-2	OPC	360	0.40	60.0	380	6.33
LC3-2	LC3	340	0.40	56.7	270	4.76
PFA-2	OPC+30% fly ash	380	0.35	54.6	300	5.49

Table 5. Secondary data from Pillai et al. (2019) [6], Table 2. The last column is derived.

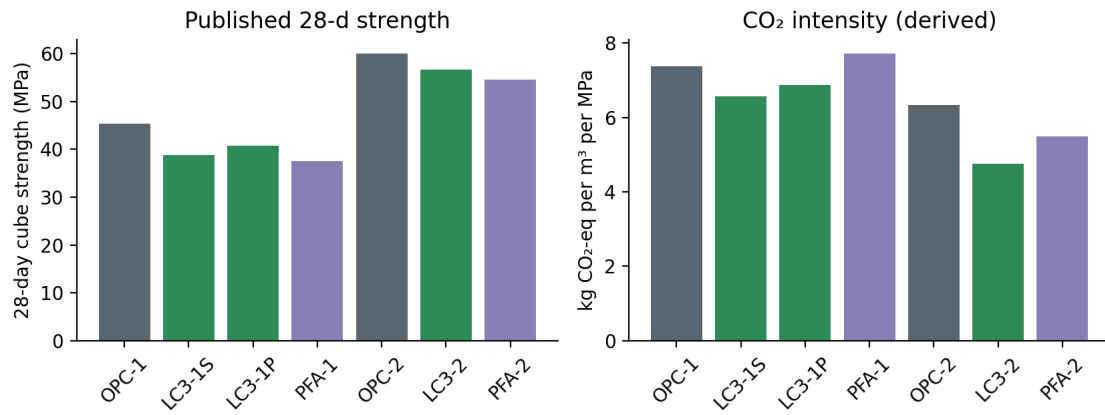


Figure 4. Published 28-day strength and derived CO₂ intensity [6].

Two points are relevant to the M25 design. First, at *identical* binder content and w/b (OPC-1 vs LC3-1S: 310 kg/m³, 0.50) the LC3 concrete reached 38.8 MPa against 45.4 MPa, about 15% lower, yet both exceeded the 35 MPa characteristic strength of their class, and the LC3 mix had the lower CO₂ per MPa. Second, when mixes are designed to the same grade, as with LC3-2 at a lower binder content than OPC-2, LC3 delivers comparable strength with about 29% less CO₂ per m³. Dhandapani et al. [7] similarly report comparable strength development for OPC and LC3 M30 and M50 concretes up to 28 days.

The mortar study of Boakye et al. [9] (Figure 5) shows the same pattern with a low-grade clay: at 50% replacement the strength deficit fell from 18% at 28 days to 5.5% at 91 days, which is consistent with continued pozzolanic reaction of the calcined clay. Their clay was lower in kaolinite than typical LC3 clays, so the 28-day deficit is likely at the high end.

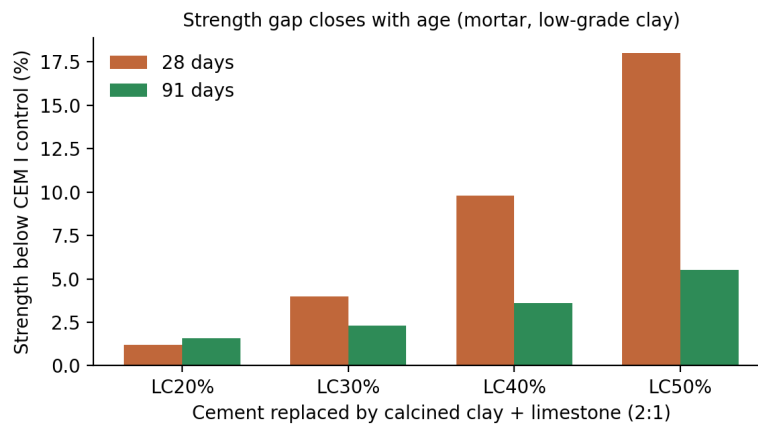


Figure 5. Strength deficit relative to a CEM I control at 28 and 91 days (mortar) [9].

5. Discussion

Equal binder is not equal strength. This study holds binder mass and w/b constant, which isolates the effect of binder chemistry on CO₂ and cost but does not guarantee that the LC3 mix meets 31.6 MPa. The published evidence suggests it is likely to, because a w/b of 0.45 is well below the 0.50 at which LC3-1S reached 38.8 MPa, but IS 10262 requires trial batches, and they are the necessary next step. If a lower w/b or more binder were required, part of the CO₂ saving would be lost.

Workability. Calcined clay raises water demand. Holding water at 186 kg/m³ implies a superplasticiser addition; the IIT Madras mixes used a PCE superplasticiser at 0.2–1% of binder mass to reach their target slump [6].

Clay quality and location. LC3 performance depends on the kaolinite content of the clay, and the CO₂ and cost advantages shrink if clay or LC3 must be transported far [6, 8]. A

location-specific assessment would be needed for any real project, including one in Jammu and Kashmir.

Durability. Although not quantified here, LC3 concretes in [6] showed chloride diffusion coefficients roughly an order of magnitude lower than OPC concretes of similar grade, which lengthens estimated service life in chloride exposure and further reduces CO₂ per year of service.

6. Limitations and future work

(1) No experimental data: specific gravities, aggregate grading and strength are assumed or taken from literature. (2) Emission factors come from a single Indian LCA and exclude transport. (3) Costs use 2016 prices and exclude processing. Future work: cast OPC and LC3 (or OPC + calcined clay-limestone blend) cubes at w/b 0.40, 0.45 and 0.50; test slump and 7/28/90-day strength; and repeat the CO₂ assessment with transport distances in an open LCA tool such as openLCA.

7. Conclusions

(i) The IS 10262 M25 OPC mix is 1 : 1.54 : 2.73 at w/c 0.45 with 413 kg/m³ cement. (ii) Replacing the binder with LC3-50 at equal mass reduces clinker from 392 to 207 kg/m³ and ground-to-gate CO₂ from 346 to 236 kg CO₂-eq/m³ (32% lower). (iii) Binder raw-material cost falls by 31–41% depending on the price of calcined clay. (iv) Published data indicate that LC3 at equal binder may show lower 28-day strength than OPC, with the gap narrowing by 90 days, so trial mixes are required to confirm that the LC3 proportions meet M25.

References

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